

INTELLIGENT RECOMMENDATION SYSTEM FOR COST EFFECTIVE PRODUCTION AND QUALITY ASSURANCE

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Dissertation submitted in partial fulfillment of the requirements for the degree of

BACHELOR OF TECHNOLOGY

BRANCH: INFORMATION TECHNOLOGY

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KNOWLEDGE DISTILLATION BASED FEAR LEVEL DETECTION IN HUMANS USING DEEP LEARNING

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SYNOPSIS

Kyda Kids, a textile company specializing in children's apparel, faces challenges in aligning production with seasonal demand fluctuations and festival-driven trends. Precise forecasting is essential to optimize inventory, reduce overproduction, and maximize sales while adapting to market dynamics.

This project introduces an AI-driven recommendation system integrating time series forecasting and sentiment analysis. The seasonal trend analysis module leverages ARIMA and SARIMA models to predict demand variations based on historical sales data, capturing seasonal spikes and festival-driven purchasing behavior. The feedback analysis module, powered by VADER-based sentiment analysis, classifies customer reviews and highlights key trends in product perception, assisting in quality assessment and strategic decision-making.

To ensure accessibility and scalability, the system is deployed as a Flask-based web application with interactive dashboards and real-time visualization of sales trends, demand forecasts, and sentiment-driven insights. The solution integrates data preprocessing pipelines, machine learning models, and automated analytics, enabling Kyda Kids to enhance production efficiency, inventory planning, and customer satisfaction through data-driven strategies.

LIST OF FIGURES

Fig. No.	Figure Name	Page No.
4.1	Schematic View of Proposed System	12
4.2	Dataset sample	14
5.1	ARIMA Sales Forecast	17
5.2	SARIMA Seasonal Forecast	18
5.3	Rolling mean and standard deviation plotted for stationarity analysis.	19
5.4	Customer Segmentation Using K-Means Clustering	20
5.5	Customer Feedback Sentiment Analysis	20
5.6	Feature Correlation	21
5.7	Flask App Home Page	22
5.8	Sales Trend Analysis	23
5.9	Feedback Analysis	24
5.10	Feedback Table	24

CONTENTS

CHAPTER	Page No.
Acknowledgement -----	(i)
Synopsis-----	(ii)
List of Figures-----	(iii)
1. INTRODUCTION-----	1
1.1 Problem Statement	1
1.2 Scope And Motivation	2
2. LITERATURE SURVEY-----	4
2.1 Related Works	4
2.2 Gaps Identified	5
3. SOFTWARE REQUIREMENT SPECIFICATION-----	7
3.1 Functional Requirements	7
3.1.1 Sales Trend Analysis	7
3.1.2 Feedback Analysis	7
3.1.3 Integrated Flask Application	7
3.2 Non-Functional Requirements	8
3.2.1 Real Time Performance	8
3.2.2 Robustness And Reliability	8
3.2.3 Scalability And Adaptability	8
3.3 Hardware Requirements	8
3.4 Software Requirements	8
4. PROPOSED SYSTEM-----	11

4.1	Proposed System	11
4.1.1	Key Objectives	11
4.1.2	ARIMA & SARIMA	11
4.1.3	Expected Benefits	11
4.2	System Workflow	12
4.3	Dataset	13
4.3.1	Data Attributes and Structure	15
5.	IMPLEMENTATION AND RESULT ANALYSIS-----	16
5.1	Data Preprocessing	16
5.1.1	Data Cleaning & Transformation	16
5.2	Model Selection and Training	17
5.2.1	ARIMA (AutoRegressive Integrated Moving Average)	17
5.2.2	SARIMA (Seasonal ARIMA) for Seasonal Forecasting	17
5.3	Stationarity and Differencing	18
5.3.1	Stationarity in Time Series	18
5.3.2	Rolling Statistics for Stationarity Analysis	18
5.3.3	Augmented Dickey-Fuller (ADF) Test for Stationarity	19
5.3.4	Differencing to achieve Stationarity	19
5.3.5	Trend and Seasonality Decomposition	19
5.4	Feedback Analysis	20
5.4.1	Customer Segmentation Using K-Means Clustering	20
5.4.2	Customer Feedback Sentiment Analysis	20
5.5	Results Of Flask App Integration	22
5.5.1	Flask App Home Page	22
5.5.2	Result-Sales Trend Analysis	23
5.5.3	Result Feedback Analysis	24
6.	CONCLUSION AND FUTURE WORK-----	26

6.1	Conclusion	26
6.2	Future Enhancement	26

BIBLIOGRAPHY-----	28
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CHAPTER 1

INTRODUCTION

Optimizing production planning in the textile industry requires a data-driven approach to adapt to seasonal demand fluctuations and festival-driven market trends. Kyda Kids, a children's apparel company, faces challenges in aligning its production strategy with changing consumer preferences, seasonal variations, and peak sales periods. Traditional forecasting methods often fail to capture complex demand patterns, leading to overproduction, stock shortages, and inefficient resource allocation.

To address this, this project introduces an AI-powered recommendation system that integrates time series forecasting and sentiment analysis for seasonal trend prediction and customer feedback analysis. The system utilizes ARIMA (AutoRegressive Integrated Moving Average) and SARIMA (Seasonal AutoRegressive Integrated Moving Average) models to analyze historical sales data and predict future demand across different seasons and festive periods. This ensures accurate production planning and efficient inventory management by identifying key sales patterns and demand spikes.

Additionally, the feedback analysis module employs VADER (Valence Aware Dictionary and Sentiment Reasoner) for Natural Language Processing (NLP)-based sentiment analysis, extracting insights from customer reviews. By categorizing sentiments into positive, neutral, or negative, the system helps assess product reception and quality issues, enabling Kyda Kids to make data-driven improvements.

To ensure accessibility and real-time data visualization, the system is deployed as a Flask-based web application, integrating interactive dashboards, predictive analytics, and automated reporting tools. This solution enables Kyda Kids to enhance production efficiency, optimize inventory planning, and improve overall business strategy through AI-driven insights.

1.1 PROBLEM STATEMENT

Kyda Kids, a children's apparel company, faces challenges in forecasting demand and optimizing production due to seasonal variations, festive trends, and evolving consumer preferences. Inefficient planning can lead to overproduction, stock shortages, increased costs, and supply chain inefficiencies, ultimately affecting profitability. Traditional forecasting methods often fail to capture complex market dynamics, making it difficult to align production with actual demand.

Additionally, manual analysis of customer feedback is time-consuming and lacks scalability, making it difficult to identify recurring product issues such as material defects, size mismatches, or design preferences. Without an automated sentiment analysis system, valuable insights into consumer preferences, emerging fashion trends, and common product complaints remain underutilized. This limits the company's ability to enhance product quality, improve customer satisfaction, and adapt to market demands effectively.

1.2 SCOPE AND MOTIVATION

SCOPE

This project aims to develop an AI-powered recommendation system to optimize seasonal demand forecasting and customer feedback analysis for Kyda Kids, a children's apparel company. The system will utilize advanced time-series forecasting models like ARIMA and SARIMA to predict seasonal trends, festive demand spikes, and purchasing patterns based on historical sales data. This will enable Kyda Kids to align production with real-time market demand, reducing inventory waste and production inefficiencies.

Additionally, the project will incorporate Natural Language Processing (NLP) techniques such as VADER sentiment analysis to extract insights from customer reviews. By categorizing feedback into positive, neutral, and negative sentiments, the system will help identify recurring product issues like fabric quality, color retention, and size mismatches. The insights will be displayed in an interactive dashboard for better decision-making.

A Flask-based web application will serve as the platform for visualizing sales trends, return analysis, and sentiment-based feedback insights. The system will provide real-time data processing, allowing decision-makers to track emerging fashion trends, consumer preferences, and market shifts, ensuring data-driven production planning and marketing strategies.

MOTIVATION

Kyda Kids operates in a highly competitive and season-dependent market, where demand fluctuations and changing consumer preferences significantly impact sales performance. Traditional forecasting methods often fail to capture complex demand patterns, leading to overproduction, excess stock, and financial losses. Additionally, a lack of structured feedback analysis limits the company's ability to identify and resolve product issues in a timely manner.

By implementing an AI-driven recommendation system, this project seeks to address these challenges through:

- Automated demand forecasting to optimize production schedules and reduce wastage.

- Data-driven inventory management to maintain the right stock levels for different seasons and festive periods.
- Real-time customer feedback analysis using NLP-based sentiment classification to enhance product design and quality.
- A dynamic web-based dashboard for seamless data visualization and decision-making.

The integration of machine learning, time-series forecasting, and sentiment analysis will empower Kyda Kids to make smarter, data-backed business decisions, increase operational efficiency, and enhance customer satisfaction, ensuring long-term growth and competitiveness in the textile industry.

CHAPTER 2

LITERATURE SURVEY

2.1 RELATED WORKS

Ref No	Title	Methodology	Limitations
1	Deep Learning for Demand Forecasting in the Fashion and Apparel Retail Industry (Goyal et al., 2023)	Deep learning models using historical sales data and product images; clustering based on sales profiles and visual similarity.	May not generalize well to unseen or rapidly changing fashion trends.
2	AI-Based Demand Forecasting for Profitability in the Retail Industry (Patel & Singh, 2022)	AI models analyze large datasets to predict demand trends, optimize inventory, and reduce excess stock.	Requires significant historical and real-time data to maintain accuracy.
3	AI Advances Precision in Fashion Demand Forecasting (Brown & Zhang, 2024)	AI-based inventory management for handling supply chain complexities and adapting to changing preferences.	Blog post, may lack experimental validation or in-depth technical details.
4	Customer Feedback Analysis using NLP (Chen et al., 2023)	Natural Language Processing to extract sentiment and trends from customer reviews.	May struggle with context understanding or slang in customer feedback.
5	The Role of Natural Language Processing in Fashion E-commerce (Gupta & Sharma, 2023)	NLP techniques to analyze sentiment and generate marketing insights; NLP-driven recommendations to enhance customer experience.	Effectiveness depends on the quality and quantity of available textual data.
6	Predicting Fashion Trends with Machine Learning (Zhang et al., 2024)	Machine learning using sales history, social media trends, and reviews to predict fashion trends and	High dependence on external data sources like social media which can be

		optimize production.	noisy or biased.
7	Time Series Forecasting in Retail (Patel & Gupta, 2023)	Time-series models (ARIMA, LSTM, Prophet) for forecasting seasonal clothing demand to reduce stockouts/overstock.	May require retraining for new seasons or trend shifts; sensitive to anomalies.
8	Sentiment Analysis in E-Commerce (Lee et al., 2023)	NLP analysis of product reviews to detect recurring complaints and enhance product quality and satisfaction.	Subjectivity in reviews and sarcasm may reduce accuracy.

2.2 GAPS IDENTIFIED

1. AI-based demand forecasting struggles with fast-changing consumer preferences, making it difficult to predict demand accurately for new and trending products. Static models often fail to capture sudden shifts influenced by social media trends and seasonal demands.
2. AI-driven demand forecasting models struggle to adjust predictions during flash sales, discounts, and economic shifts. Since these models rely on stable pricing trends, they often fail to capture short-term spikes in demand, leading to inventory mismanagement.
3. NLP-based feedback analysis misses detailed product issues (e.g., fabric quality, size mismatch) and fails to turn insights into actionable recommendations. Sentiment classification alone does not provide specific defect patterns that could help improve products.
4. Forecasting models struggle with seasonal demand shifts, leading to inventory mismanagement and ignoring regional and cultural influences. This results in overproduction of low-demand items and shortages of high-demand seasonal products.
5. AI-based recommendations often fail to identify which products are frequently bought together, leading to ineffective bundling strategies. Without analyzing contextual buying habits, businesses miss the chance to maximize cross-selling opportunities.
6. AI models are static and do not dynamically adjust to real-time sales trends, making predictions outdated. Once trained on historical data, they fail to incorporate emerging consumer behavior changes.
7. Return analysis models are biased towards common issues (e.g., size) while overlooking critical but less frequent problems (e.g., fabric defects). This leads to

manufacturers prioritizing easily noticeable issues while neglecting less-reported but impactful concerns.

8. Long-term fashion trend prediction is inaccurate, as models fail to anticipate macroeconomic and consumer behavior shifts over time. External factors like economic downturns and sustainability trends are often not integrated into forecasting models.
9. Existing models fail to incorporate external factors like weather, local festivals, and economic shifts, reducing recommendation accuracy. Consumer demand often fluctuates due to these influences, but static models cannot adapt in real time, leading to missed sales opportunities.
10. Most AI models prioritize historical trends while ignoring real-time consumer shifts, such as viral trends or influencer promotions. This limits businesses from capitalizing on sudden demand surges, resulting in lost revenue opportunities.
11. AI systems process customer feedback in batches rather than in real time, causing delays in addressing product issues. Businesses may take weeks to react to emerging concerns, reducing customer satisfaction and retention.

CHAPTER 3

SOFTWARE REQUIREMENTS SPECIFICATIONS

System requirements refer to the minimum and maximum hardware and software specifications necessary for a system or application to function correctly. To accurately determine these requirements, it is essential to thoroughly analyze the functional and non-functional needs of the system, as well as to identify any constraints or factors that could impact its performance. This process ensures that the system operates efficiently within its intended environment, providing optimal user experience and functionality.

3.1 FUNCTIONAL REQUIREMENTS

3.1.1 Sales Trend Analysis

The system must analyze historical sales data to identify trends and recommend products based on seasonal demand and festival-based purchasing patterns. It should allow users to search by Product ID and provide recommendations for both individual products and grouped items. The forecasting models (ARIMA, SARIMA) should adapt dynamically to market shifts.

3.1.2 Feedback Analysis

The system must analyze customer feedback using VADER (Valence Aware Dictionary and sEntiment Reasoner) to classify reviews as Positive, Negative, Neutral, or Mixed. If feedback is classified as Negative or Mixed, the system must extract and display the primary reasons behind dissatisfaction, such as size issues, fabric defects, or color fading. This enables Kyda Kids to identify and address recurring product concerns effectively.

3.1.3 Integrated Flask Application

The system must integrate Sales Trend Analysis and Feedback Analysis into a colorful, user-friendly Flask web application. The application should allow users to:

- **Search for a Product ID** and receive recommendations based on seasonal trends and festivals.
- **Analyze Customer Feedback**, displaying sentiment classifications (Positive, Negative, Neutral, Mixed) along with reasons for dissatisfaction.
- **Visualize Insights with Graphs & Charts**, making it easier for Kyda Kids employees to interpret trends and take action.
- **Ensure Real-Time Data Updates**, dynamically reflecting the latest product sales and customer sentiment data.

3.2 NON-FUNCTIONAL REQUIREMENTS

3.2.1 Real-Time Performance

The system must deliver real-time responsiveness in Sales Trend Analysis and Feedback Analysis, ensuring minimal processing delays. The Flask application should dynamically update product recommendations and sentiment insights, allowing Kyda Kids to make timely business decisions.

3.2.2 Robustness and Reliability

The system should function reliably across varied datasets and user queries, ensuring consistent recommendations and feedback insights. It must handle diverse input sources, including large-scale sales data and customer reviews, without compromising accuracy. Additionally, the application should remain stable under high data loads and multiple user interactions.

3.2.3 Scalability and Adaptability

The system should be scalable, supporting an expanding dataset of product sales and customer feedback over time. It must be adaptable to new seasonal trends, festivals, and customer preferences. The Flask application should also be designed to integrate future enhancements, such as additional machine learning models or new data visualization techniques, ensuring long-term usability.

3.3 HARDWARE REQUIREMENTS

- GPU: NVIDIA GeForce GTX or AMD Radeon RX series.
- Graphics Card: RTX 3050.
- Storage: Minimum 20 GB free disk space.
- RAM: 16 GB
- OS: Windows 10 or above
- Processor: 64-bit processor

3.4 SOFTWARE REQUIREMENTS

- Google Colaboratory
- Numpy
- Pytorch
- Tensorflow 2.16.2
- Matplotlib
- Pandas
- Scikit-learn
- Plotly
- VADER

- Flask
- CSV

Google Colaboratory

Google Colaboratory is used for developing and testing machine learning models, such as ARIMA, SARIMA, and sentiment analysis with VADER. It provides a cloud-based environment for data preprocessing, model training, and experimentation without requiring local computational resources.

NumPy

NumPy is essential for handling numerical operations in sales trend forecasting and feedback analysis. It enables efficient computation of time series data, statistical metrics, and array manipulations required for generating accurate recommendations.

PyTorch

PyTorch is used for deep learning-based improvements in recommendation and sentiment analysis models. It supports the potential integration of advanced neural networks for enhanced product trend prediction and customer feedback classification.

TensorFlow 2.16.2

TensorFlow is utilized for building and optimizing machine learning models in the Kyda Kids recommendation system. It aids in training ARIMA-based forecasting models and refining sentiment classification algorithms for better decision-making.

Matplotlib

Matplotlib is employed for visualizing sales trends, seasonal variations, and customer feedback insights in the Flask application. It helps generate line charts, bar graphs, and distribution plots to present data in an intuitive manner.

Pandas

Pandas is crucial for data preprocessing and analysis, enabling efficient loading, filtering, and structuring of sales and feedback datasets. It ensures smooth handling of large CSV files used in trend analysis and customer sentiment evaluation.

Scikit-learn

Scikit-learn provides machine learning utilities for feature extraction, data normalization, and model evaluation in Kyda Kids' recommendation and feedback analysis. It is also used for implementing classification and clustering techniques.

Plotly

Plotly is integrated for interactive data visualization in the Flask application, allowing users to explore seasonal sales trends, product demand patterns, and customer sentiment analytics through dynamic charts.

VADER

VADER (Valence Aware Dictionary and sEntiment Reasoner) is utilized for analyzing customer feedback sentiment. It classifies reviews into positive, neutral, negative, and mixed sentiments, helping Kyda Kids identify product improvement areas.

Flask

Flask is used for developing the Kyda Kids web application, integrating sales trend forecasting and sentiment analysis modules into a user-friendly interface. It enables dynamic updates based on real-time data processing.

CSV

CSV files serve as the primary data storage format for historical sales and customer feedback. They facilitate efficient data retrieval, processing, and analysis in the Kyda Kids recommendation system.

CHAPTER 4

PROPOSED SYSTEM

4.1 PROPOSED SYSTEM

The proposed system aims to enhance seasonal sales forecasting for Kyda Kids, a textile company specializing in children's clothing. By leveraging time series forecasting techniques (ARIMA & SARIMA), the system enables data-driven decision-making for optimizing inventory management, production planning, and sales strategies.

4.1.1 Key Objectives

1. Accurately forecast future sales trends based on historical data.
2. Capture seasonal demand variations to avoid stock shortages and overproduction.
3. Improve business efficiency by optimizing supply chain and marketing strategies.
4. Provide insights into best- and worst-performing products for different seasons.

4.1.2 ARIMA & SARIMA

- ARIMA is effective for trend-based forecasting where seasonal patterns are not dominant.
- SARIMA extends ARIMA by incorporating seasonal components, making it ideal for products with fluctuating demand across seasons.
- Compared to deep learning models (LSTMs, RNNs), ARIMA/SARIMA performs well on structured time-series data with fewer computational requirements

4.1.3 Expected Benefits

1. Improved forecasting accuracy → Better stock management
2. Reduced return rates → Minimized overproduction losses
3. Enhanced decision-making → Optimized sales & marketing strategies

4.2 SYSTEM WORKFLOW

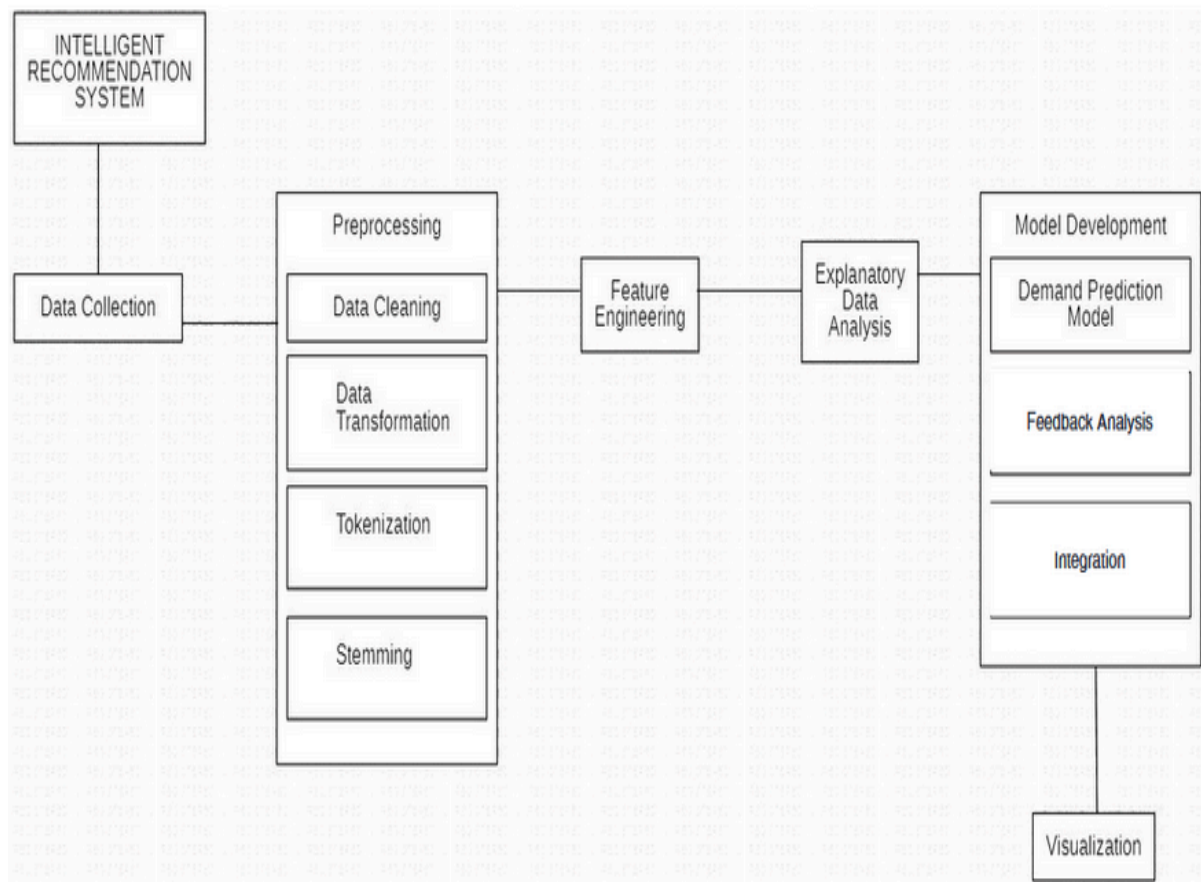


Fig 4.1 Schematic view of proposed system

The given diagram outlines the architecture of an intelligent recommendation system, and it can be effectively associated with our seasonal forecasting and recommendation system for Kyda Kids. Our project, which focuses on optimizing production based on historical sales data, seasonal trends, and customer return patterns, aligns well with the workflow depicted in the image.

At the core of our system is data collection, which gathers historical sales data, customer purchase behaviors, product metadata (such as material composition, styles, and seasonal demand), and return statistics. This step ensures that the model has a rich dataset to analyze before making forecasts. The collected data is then subjected to preprocessing, where data cleaning eliminates missing values, outliers, and duplicate entries. Data transformation normalizes numerical attributes, encodes categorical features (e.g., product types and sizes), and scales values to ensure consistency across different sources. If customer reviews or textual feedback are included, tokenization and stemming will be

applied to process textual data, making it usable for NLP-based sentiment analysis, which can further refine demand predictions.

Once preprocessed, the data undergoes feature engineering, where meaningful insights are extracted. For instance, seasonal attributes such as winter or summer collections, fabric types (fleece, polyester, spandex), and previous demand trends are transformed into key features that impact future sales. These engineered features are then used in explanatory data analysis (EDA), where patterns are explored visually through correlation matrices, histograms, and trend lines. This helps in identifying key sales trends, such as which clothing materials perform best in specific seasons and whether return rates are influenced by material quality or sizing inconsistencies.

Moving forward, the model development phase focuses on demand prediction using ARIMA and SARIMA. These models learn from historical sales patterns to forecast demand for upcoming seasons, ensuring that Kyda Kids produces an optimal quantity of products while minimizing excess inventory. Additionally, a feedback analysis mechanism evaluates customer return reasons and purchase satisfaction scores, helping to refine future predictions. This integrates seamlessly with our recommendation system, which suggests the best product categories to manufacture based on predicted demand.

Once trained, the model's performance is validated through accuracy metrics such as RMSE (Root Mean Square Error) and MAPE (Mean Absolute Percentage Error). If the model does not meet the desired accuracy threshold, further fine-tuning and cross-validation techniques are applied to enhance its predictive capability. Finally, the visualization component plays a crucial role in presenting these insights through a user-friendly Flask-based interface. This interface displays key performance indicators (KPIs), best-performing products for each season, forecasted sales figures, and recommended production adjustments, allowing stakeholders to make data-driven decisions efficiently.

4.3 DATASET

The dataset plays a crucial role in training and evaluating the ARIMA and SARIMA models for sales forecasting. Below are the key aspects of the dataset used in this project: The dataset consists of **10,000+ entries** and contains sales-related information over a period of time. It includes various attributes that capture product characteristics, sales figures, seasonality, and customer behavior.

Product_ID	Product_Type	Material	Color	Sales_Data	Return_Rate	Production_Data	Season	Gender	Season_Demand_Spike	Date	Age	Customer_Feedback
P0166	T-Shirt	Fleece	Pink	231	0.17	325	Winter	Unisex	1.45	2023-01-01	13	Feels breathable and light.
P0058	T-Shirt	Cotton	White	285	0.05	319	Winter	Male	1.22	2023-01-01	5	Loved the design and comfort.
P0099	Jacket	Cotton	Grey	345	0.26	389	Winter	Unisex	1.13	2023-01-01	10	Loved the design and comfort.
P0087	Jacket	Cotton	Green	64	0.3	196	Winter	Male	1.3	2023-01-01	9	The stitching quality could be better.
P0096	Jacket	Cotton	White	48	0.06	444	Winter	Unisex	1.41	2023-01-01	9	The stitching quality could be better.
P0146	Hoodie	Fleece	Red	122	0.14	179	Winter	Male	1.31	2023-01-01	9	The color is vibrant and attractive.
P0143	Sweatpants	Fleece	Yellow	81	0.25	311	Winter	Male	1.3	2023-01-01	4	The fabric is soft and comfortable.
P0124	Jacket	Polyester	Blue	300	0.17	310	Winter	Male	1.11	2023-01-01	7	The fabric is soft and comfortable.
P0032	Sweatpants	Polyester	White	120	0.07	166	Winter	Unisex	1.11	2023-01-01	14	The fabric is soft and comfortable.
P0047	Leggings	Wool	Green	6	0.07	335	Winter	Unisex	1.34	2023-01-01	9	The fabric is soft and comfortable.
P0184	Sweatpants	Cotton	Green	23	0.14	390	Winter	Unisex	1.43	2023-01-01	11	The stitching quality could be better.
P0089	Sweatpants	Cotton	Pink	235	0.25	406	Winter	Male	1.33	2023-01-01	11	The stitching quality could be better.
P0095	Jacket	Cotton	Red	83	0.29	137	Winter	Male	1.45	2023-01-01	5	Perfect for summer wear.
P0092	Sweatpants	Fleece	Yellow	161	0.27	244	Winter	Male	1.42	2023-01-01	14	The product arrived damaged.

Fig 4.2 Dataset sample

Key Attributes in the Dataset:

Product_ID	Unique identifier for each product.
Product_Type	Category of the product (e.g., T-shirts, Jackets, Pants).
Material	Fabric type (e.g., Polyester, Cotton, Fleece).
Color	Product color variations.
Sales_Data	Number of units sold over time.
Return_Rate	Percentage of products returned by customers.
Production_Data	Number of products manufactured per batch.
Season	The season when the product was primarily sold.
Gender	The season when the product was primarily sold.
Season_Demand_Spike	Whether there is a demand spike in specific seasons.
Age	Age group of the customers purchasing the product.
Customer_Feedback	Age group of the customers purchasing the product.

4.3.1 Data Attributes and Structure

Each row in the dataset corresponds to a single purchase instance, with a timestamped sales record indicating when the product was bought. This timestamp is crucial for time-series forecasting models like ARIMA and SARIMA, as it enables us to observe long-term trends, seasonal cycles, and periodic fluctuations in customer purchasing behavior. The product category field helps classify sales based on different clothing types such as jackets, sweaters, t-shirts, and pants, while the material type column (e.g., polyester, fleece, spandex) provides insight into fabric preferences across different weather conditions.

The sales quantity and price per unit allow us to calculate revenue trends and assess whether certain product categories are more profitable than others. Additionally, the customer ID field enables segmentation, helping us analyze buying patterns among different consumer groups, such as repeat customers vs. first-time buyers. The return status column is a critical aspect of our dataset, as it helps identify common issues that lead to product returns, whether due to material quality, sizing mismatches, or seasonal demand misalignment.

A key aspect of the dataset is its seasonal nature, meaning sales patterns fluctuate depending on the time of year. For example, winter-related products like fleece jackets and wool sweaters tend to sell more during colder months, whereas lighter clothing like cotton t-shirts and shorts experience higher demand in summer. This seasonal variation is captured using monthly aggregation of sales data, a preprocessing step necessary for SARIMA-based forecasting, which explicitly models both seasonality and trends.

Another important observation is the impact of promotions and discounts on sales performance. Certain entries in the dataset show spikes in sales volume, particularly during festive seasons or major sales events like Black Friday or end-of-season clearance sales. These external factors must be accounted for when training forecasting models to avoid overfitting to short-term anomalies.

CHAPTER 5

IMPLEMENTATION AND RESULT ANALYSIS

The implementation of the Kyda Kids Seasonal Forecasting System is structured into multiple stages to ensure accurate sales forecasting and optimal production planning. The development process involved data collection, preprocessing, time-series modeling using ARIMA and SARIMA, system integration, and deployment via a Flask-based GUI. The key goal of the system is to provide accurate seasonal recommendations for Kyda Kids' children's clothing products, reducing overproduction and optimizing inventory management.

5.1 DATA PREPROCESSING

Before developing the forecasting model, an extensive dataset was prepared using historical sales data from Kyda Kids. The dataset contained **10,000+ entries**, covering various product categories such as fleece, polyester, and spandex-based clothing items. The raw dataset was structured as follows:

- **Date:** The sales date for each transaction.
- **Product ID:** Unique identifier for each clothing item.
- **Category:** Type of clothing (e.g., winter wear, summer outfits).
- **Sales Quantity:** Number of units sold.
- **Return Data:** Number of items returned and reasons for return.
- **Price Information:** Cost per unit, discounts applied.

5.1.1 Data Cleaning & Transformation

To ensure data quality, the following preprocessing steps were applied:

→ **Handling Missing Data:**

- ◆ Sales data with missing values were filled using forward-fill interpolation.
- ◆ Missing price values were replaced with the mean of similar products in the same category.

→ **Date Standardization & Aggregation:**

- ◆ The Date column was converted to datetime format to facilitate time-series analysis.
- ◆ Data was resampled to a monthly level, aggregating total sales per month for each product category.
- ◆ This monthly aggregation is crucial because ARIMA/SARIMA models require evenly spaced time-series data for effective forecasting.

→ Feature Engineering:

- ◆ **Seasonal Indicators:** A new column was added to denote whether the sale occurred in winter, summer, spring, or autumn.
- ◆ **Rolling Averages:** Moving averages were calculated to smooth out short-term fluctuations.
- ◆ **Trend Decomposition:** Sales trends were extracted to separate seasonal effects from irregular variations.

5.2 MODEL SELECTION AND TRAINING

To accurately predict future sales, we experimented with two time-series forecasting models:

5.2.1 ARIMA (AutoRegressive Integrated Moving Average)

- ARIMA requires the dataset to be stationary, meaning that the mean and variance should remain constant over time.
- Stationarity was checked using the Augmented Dickey-Fuller (ADF) test.
- Differencing (d parameter) was applied to remove trends and ensure stationarity.
- The optimal (p, d, q) values were determined using Auto ARIMA (pmdarima).
- The final ARIMA model was trained using the (3, 1, 3) configuration, which provided the lowest error rate.

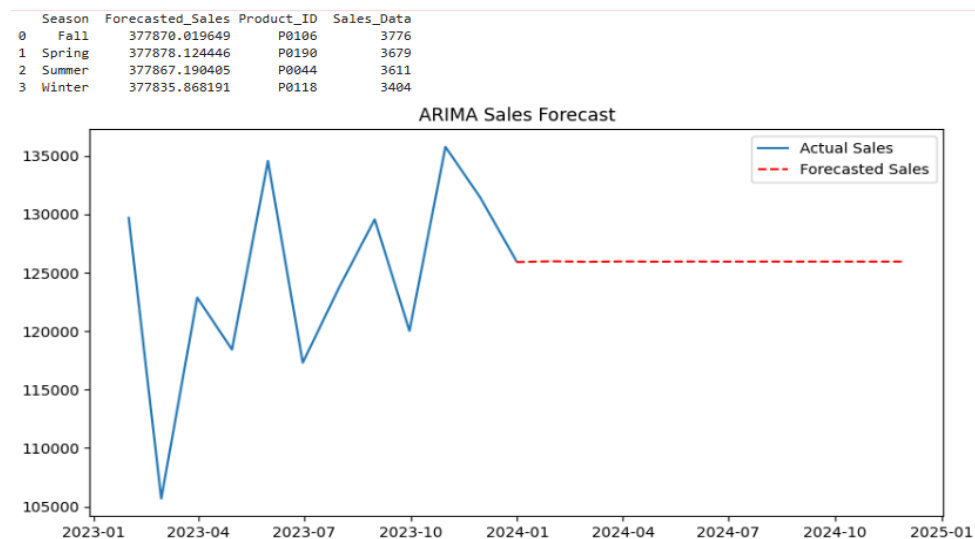


Fig. 5.1 ARIMA Sales Forecast

5.2.2 SARIMA (Seasonal ARIMA) for Seasonal Forecasting

Since sales patterns exhibit strong seasonal trends, **SARIMA** was implemented:

- Unlike ARIMA, SARIMA can handle seasonal fluctuations using additional parameters (P, D, Q, s) , where:
 1. P = Seasonal autoregressive term.
 2. D = Seasonal differencing to remove seasonal trends.
 3. Q = Seasonal moving average term.
 4. s = Seasonal period (set to 12 for monthly seasonality).
- Grid search was performed to find the best $(p, d, q) \times (P, D, Q, s)$ combination.
- The final SARIMA model $(2, 1, 2) \times (1, 1, 1, 12)$ was selected for deployment.

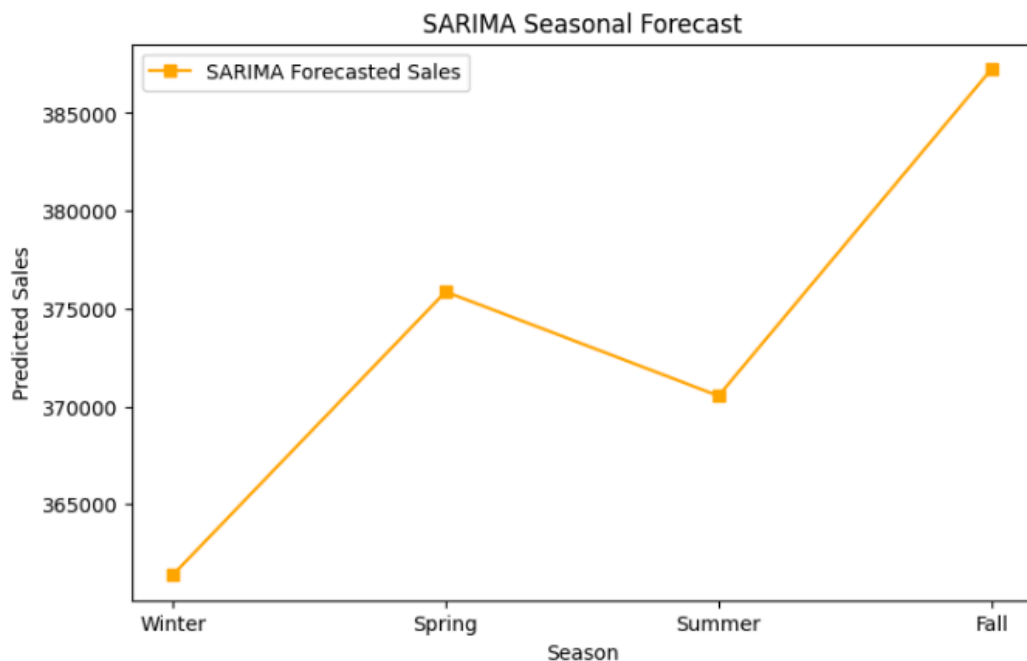


Fig. 5.2 SARIMA Seasonal Forecast

5.3 STATIONARITY AND DIFFERENCING

5.3.1 Stationarity in Time Series

- Definition and importance of stationarity
- Types of stationarity:
 - Strict stationarity (constant mean and variance)
 - Weak stationarity (only mean and variance are constant)
- Effects of non-stationarity on forecasting models

5.3.2 Rolling Statistics for Stationarity Analysis

- Rolling Mean: Moving average over a defined window (e.g., 12 months)
- Rolling Standard Deviation: Measures variability within the rolling window
- Interpreting Rolling Statistics:

- If the rolling mean and standard deviation are stable, the series is stationary
- If they fluctuate significantly, the series is non-stationary

5.3.3 Augmented Dickey-Fuller (ADF) Test for Stationarity

- Hypothesis Testing:
 - Null Hypothesis (H_0): The series is non-stationary
 - Alternative Hypothesis (H_1): The series is stationary
- p-value interpretation:
 - If $p < 0.05$: Reject $H_0 \rightarrow$ Series is stationary
 - If $p > 0.05$: Fail to reject $H_0 \rightarrow$ Series is non-stationary

5.3.4 Differencing to Achieve Stationarity

- First-order differencing to remove trends
- Second-order differencing if first-order is not sufficient
- Seasonal differencing to remove seasonal effects

5.3.5 Trend and Seasonality Decomposition

- Additive decomposition: Used when trend and seasonality have a constant magnitude
- Multiplicative decomposition: Used when trend and seasonality grow proportionally
- Implementation using `statsmodels.tsa.seasonal_decompose()`

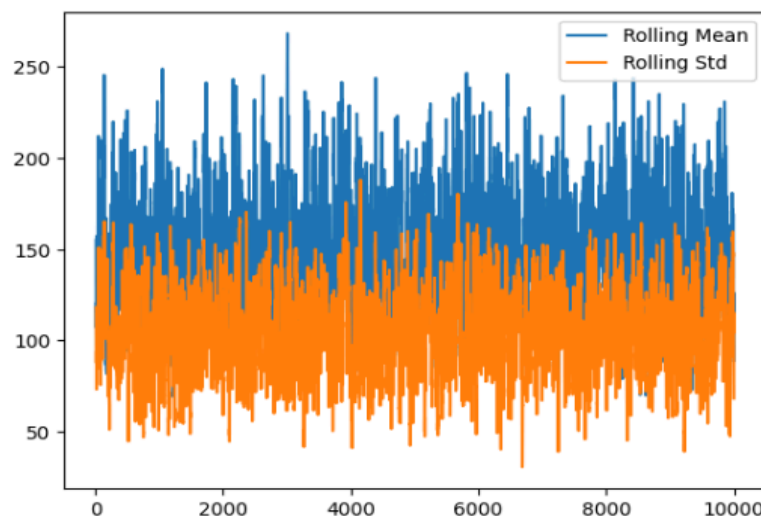


Fig. 5.3 Rolling mean and standard deviation plotted for stationarity analysis.

5.4 FEEDBACK ANALYSIS

5.4.1 Customer Segmentation Using K-Means Clustering

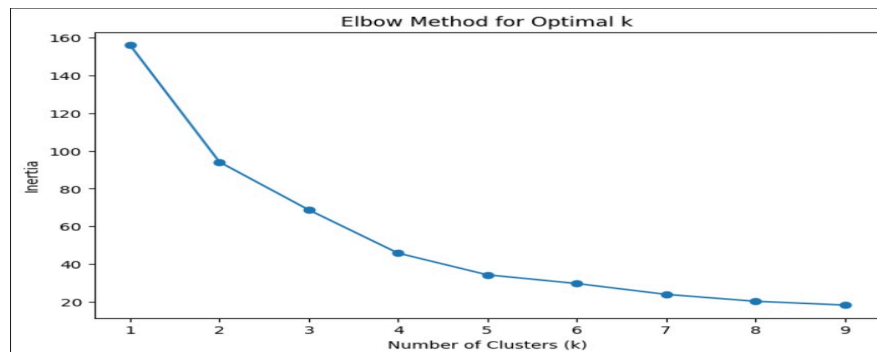


Fig. 5.4 Customer Segmentation Using K-Means Clustering

From figure 5.9, To enhance product targeting and understand customer behavior, K-Means clustering was applied to segment customers based on key purchasing features. The Elbow Method, as shown in the figure above, was used to determine the optimal number of clusters (k). This technique plots the number of clusters against the within-cluster sum of squares (inertia). From the graph, the point at which the inertia starts to decrease more slowly—forming an “elbow”—indicates the optimal k value. In this case, the elbow appears at **k=4**, suggesting that dividing the dataset into four clusters would provide a good balance between granularity and interpretability. These clusters represent distinct customer groups with similar purchasing patterns or preferences. Segmenting customers this way allows the business to tailor marketing strategies, personalized recommendations, and improve inventory planning by focusing on specific group behaviors. This implementation demonstrates how unsupervised learning techniques can provide valuable insights for data-driven decision-making in retail environments like Kyda Kids.

5.4.2 Customer Feedback Sentiment Analysis

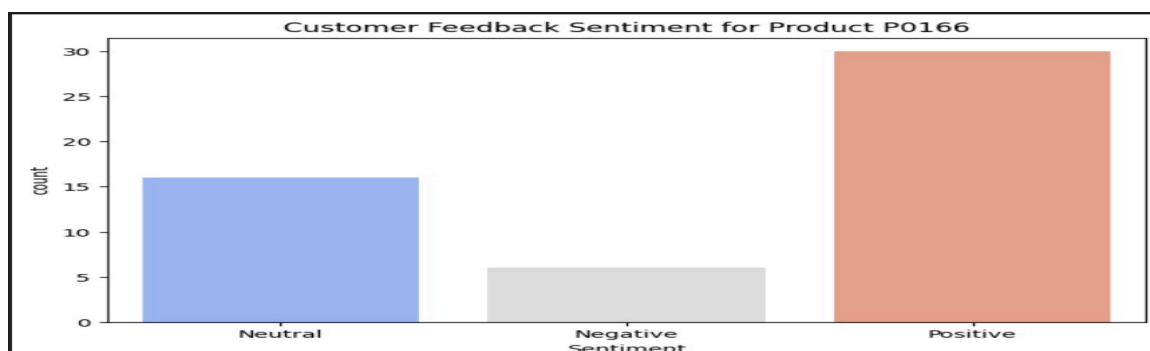


Fig. 5.5 Customer Feedback Sentiment Analysis

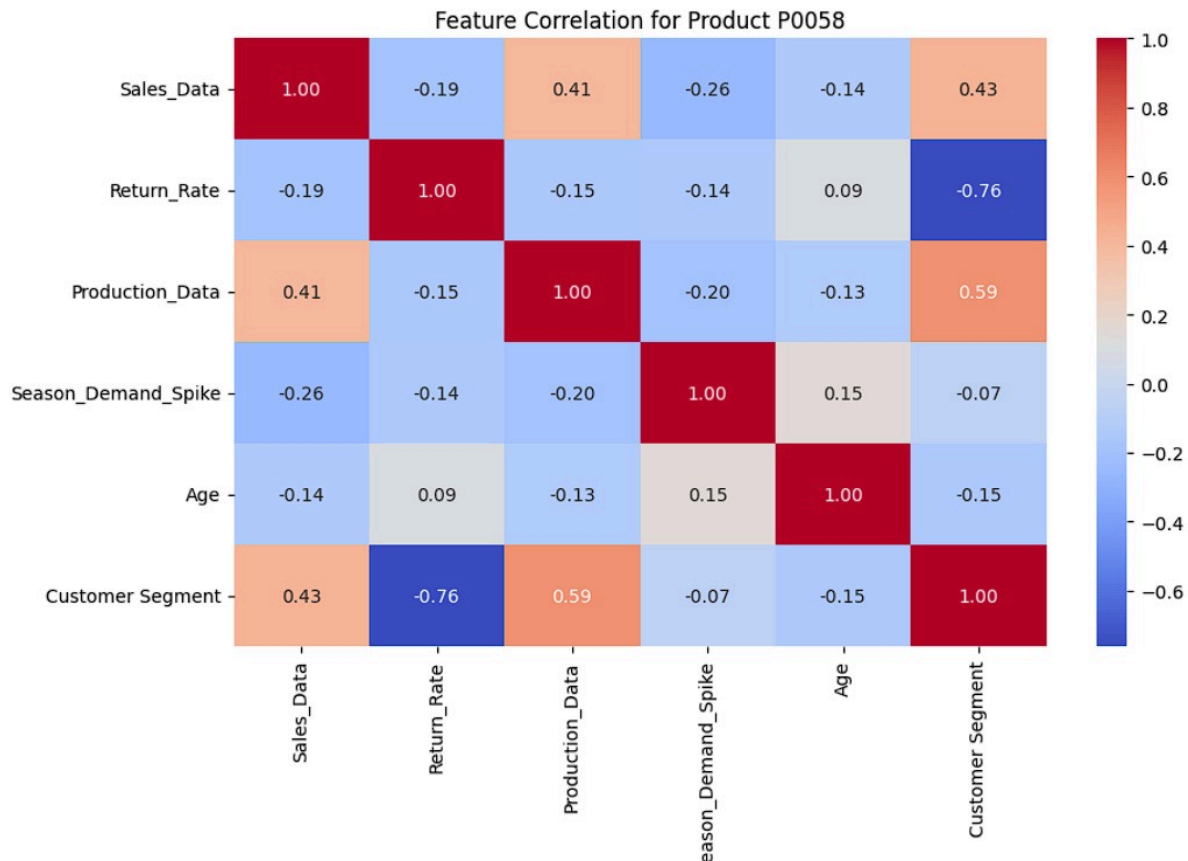


Fig 5.6 Feature Correlation

From figure 5.7, Customer feedback sentiment analysis was conducted to evaluate the reception of Product P0166 using NLP-based sentiment classification. The bar chart illustrates the distribution of customer sentiments into three categories: Positive, Neutral, and Negative. A majority of customers expressed positive feedback, indicating a high level of satisfaction with the product. Neutral feedback represents customers who had a mixed or moderate experience, while a smaller portion of reviews were negative, highlighting potential issues.

In addition to sentiment analysis, feature correlation analysis was performed to determine the impact of various product attributes (e.g., material, size, durability) on customer sentiment. This helps identify which factors contribute to positive or negative reviews. For instance, if fabric quality has a strong correlation with negative feedback, improvements in material selection can enhance customer satisfaction. By integrating sentiment trends with feature correlation insights, businesses like Kyda Kids can refine product quality, optimize inventory decisions, and develop targeted marketing strategies to enhance overall customer engagement.

5.5 RESULTS OF FLASK APP INTEGRATION

A Flask web application was developed to integrate the Kyda Kids recommendation system, enabling employees to analyze sales trends, customer feedback, and returns for better inventory and product recommendations. The backend, built with Flask, serves as the API layer for recommendation models. The frontend, designed with HTML, CSS, and JavaScript, ensures a user-friendly experience. The system leverages ARIMA and SARIMA models for sales forecasting and collaborative filtering for product recommendations. The NLP-based feedback module classifies reviews as positive, neutral, or negative, providing insights into customer satisfaction. Interactive charts, graphs, and dashboards enhance decision-making. The Flask app retrieves real-time data, helping Kyda Kids optimize sales, reduce returns, and improve product quality.

5.5.1 FLASK APP HOME PAGE



FIG 5.7 FLASK APP HOME PAGE

The home page of the Kyda Kids recommendation system serves as the main interface, providing access to two key modules: Sales Trend Analysis and Feedback Analysis. The Sales Trend Analysis section enables users to explore historical and seasonal sales trends using ARIMA and SARIMA models for demand forecasting. Users can enter a Product ID to analyze its sales performance and receive recommendations for similar products based on material, type, and color. To enhance data interpretation, line charts and bar graphs visually represent sales patterns.

The Feedback Analysis module focuses on processing customer reviews using NLP-based sentiment classification, categorizing them as positive, neutral, or negative. It also identifies common product issues such as size mismatch or fabric defects, offering actionable insights for product improvement and return reduction. The analysis results are displayed using interactive visualizations, allowing businesses to quickly assess customer

satisfaction. Designed for a seamless user experience, the home page ensures smooth navigation between modules, helping businesses make data-driven decisions efficiently.

5.5.2 RESULT-SALES TREND ANALYSIS



Fig 5.8 Sales Trend Analysis

The Sales Trend Analysis module provides a comprehensive view of product demand patterns over time, helping Kyda Kids optimize inventory and production strategies. The displayed graph represents historical sales trends, identifying seasonal fluctuations and demand spikes. By leveraging ARIMA and SARIMA models, the system forecasts future sales, allowing businesses to anticipate demand changes and reduce stockouts or overproduction.

Alongside the sales trend visualization, the module generates personalized product recommendations based on key attributes such as material, type, and color. When a user inputs a Product ID, the system analyzes past sales performance and suggests similar products that align with market demand. These recommendations assist in inventory planning and ensure better customer satisfaction by offering alternative products that match consumer preferences. The combination of trend analysis and intelligent recommendations empowers businesses to make data-driven decisions and enhance overall sales performance.

5.5.3 RESULT FEEDBACK ANALYSIS

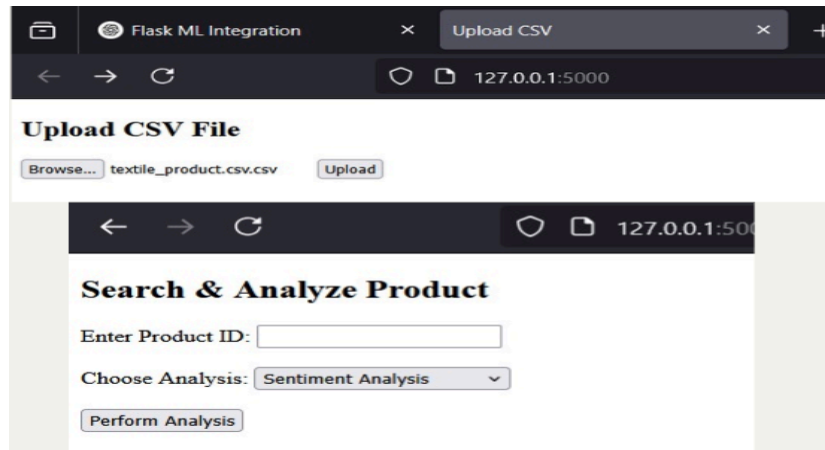


Fig 5.9 Feedback analysis

The Feedback Analysis Module processes customer reviews from the uploaded CSV file, extracting insights to assess product performance and customer satisfaction. Once the file is uploaded, the system performs sentiment analysis using NLP techniques, classifying reviews into positive, neutral, or negative categories. This helps Kyda Kids understand consumer opinions, identify recurring issues such as fabric defects, sizing mismatches, or color fading, and improve product quality accordingly.

Product ID	Product Type	Color	Material	Sentiment	Action
P0171	Hoodie	Black	Fleece	Positive	No Action Needed
P0171	Hoodie	Blue	Fleece	Positive	No Action Needed
P0171	Hoodie	Green	Fleece	Positive	No Action Needed
P0171	Hoodie	Grey	Fleece	Positive	No Action Needed
P0171	Hoodie	Pink	Fleece	Positive	No Action Needed
P0171	Hoodie	Red	Fleece	Negative	Show Reasons Areas to Improve: Product Damage
P0171	Hoodie	White	Fleece	Mixed	Show Reasons Areas to Improve: Color Fade

FIG 5.10 FEEDBACK TABLE

After performing sentiment analysis for the searched individual product, a table is displayed showing the sentiment classification for each customer review—categorized as positive, neutral, or negative. This table provides a detailed breakdown of customer perceptions, helping businesses understand product reception at a granular level.

An "Action" button is available next to each product entry. When clicked, it reveals the specific reasons behind negative and mixed feedback, such as fabric quality issues, sizing mismatches, or durability concerns. This feature allows businesses to pinpoint and address critical product shortcomings, improving overall customer satisfaction and reducing return rates.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 CONCLUSION

The Kyda Kids recommendation system successfully integrates sales trend analysis and feedback-based insights to enhance product recommendations and inventory management. By leveraging AI-driven demand forecasting, the system identifies seasonal trends and optimizes stock levels to reduce overproduction and stockouts. The feedback analysis module provides deeper insights into customer sentiment, allowing the identification of specific product issues such as fabric quality, sizing mismatches, and durability concerns.

The seamless integration of these features enables Kyda Kids to make data-driven decisions, improving product quality and customer satisfaction while minimizing return rates. Additionally, the interactive UI allows for efficient product searches and personalized recommendations, enhancing the overall user experience. The implementation of this system demonstrates the practical impact of AI and data analytics in the retail industry, paving the way for future enhancements such as real-time updates, dynamic pricing strategies, and advanced trend forecasting. This project serves as a foundation for further research and development in optimizing recommendation systems for fashion retail.

6.2 FUTURE ENHANCEMENT

Future enhancements for the Kyda Kids Recommendation System can focus on improving accuracy, adaptability, and user experience by incorporating advanced AI techniques and additional data sources. One major improvement is the integration of real-time sales and customer feedback data, allowing the system to dynamically update recommendations based on current market trends and customer preferences. By implementing reinforcement learning-based recommendation algorithms, the system can continuously refine its suggestions based on real-time interactions and purchasing patterns.

Expanding the feedback analysis module to incorporate Natural Language Processing (NLP) techniques like aspect-based sentiment analysis (ABSA) can help extract more precise insights regarding product quality, durability, and design preferences. Additionally, integrating image-based analysis using computer vision can enhance product

recommendations by analyzing customer-uploaded images of returned or defective items, allowing for automated quality assessment and design improvements.

Another key area for enhancement is personalized recommendations using deep learning models, which can analyze individual purchase histories and browsing behavior to provide tailored product suggestions. Expanding the dataset by incorporating social media trends, weather patterns, and regional preferences can further refine the accuracy of seasonal and festival-based product recommendations.

To improve business decision-making, the system can be upgraded with predictive analytics for inventory optimization, ensuring that Kyda Kids maintains optimal stock levels while reducing wastage and overproduction. Implementing a user-friendly dashboard with interactive visualizations will allow employees to analyze trends and insights more effectively.

Future iterations can also focus on multi-platform integration, enabling the system to be accessible across different digital channels, including e-commerce websites, mobile apps, and in-store kiosks. These advancements will make the Kyda Kids recommendation system more adaptive, intelligent, and efficient, leading to better customer engagement and optimized business operations.

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